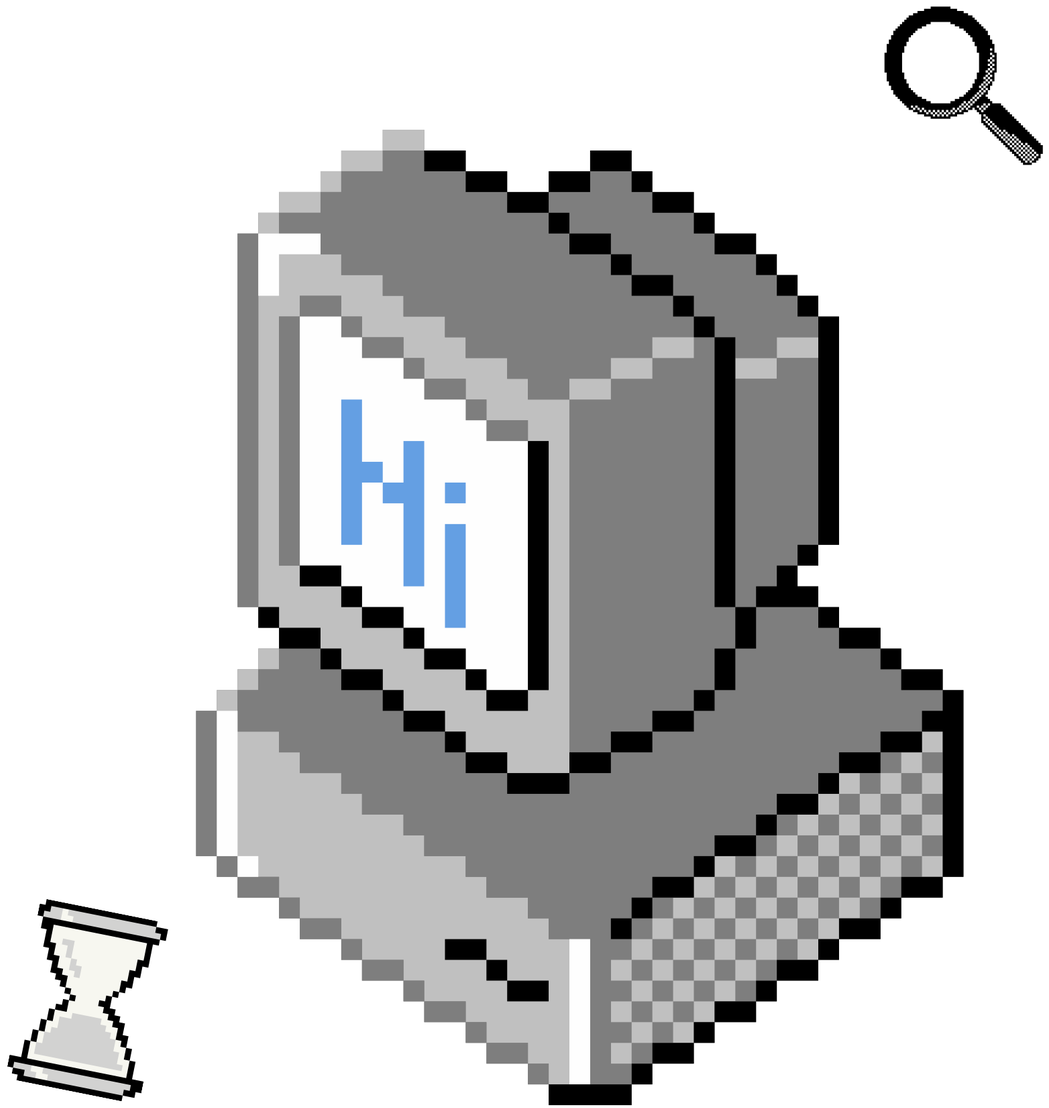


應用於室內場域之

邊緣AI去識別化監控系統

Edge AI De-identification Surveillance System for Indoor Environments

S1354001-劉羿宗

S1354019-阮玉達

S1354042-陳詠欣

S1354048-張語珊

CONTENTS

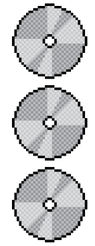
1 Problems We Identified

2 Proposed Solution

3 System Block Diagram

4 Use Case Demonstration

5 Project Timeline



Problems We Identified

Ghost Bookings & Resource Waste

Frequent no-shows and unattended belongings cause low physical utilization.

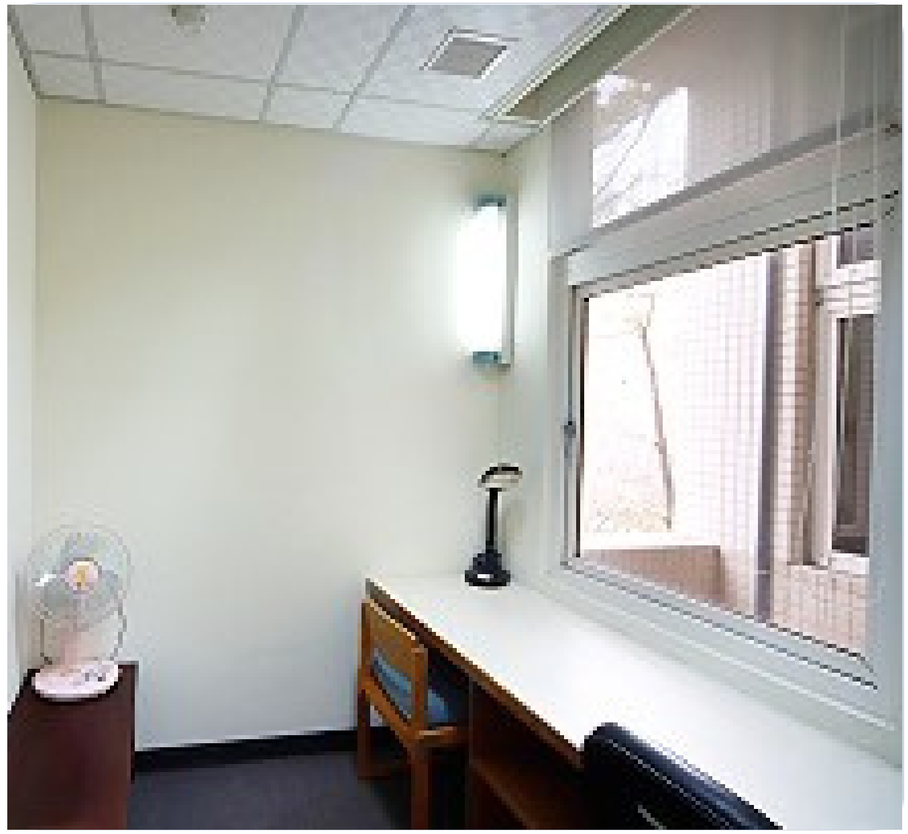
Heavy Manual Patrol Overhead

Door-to-door inspections drain staff labor and fail to provide 24/7 oversight.

Privacy Concerns & Legal Constraints

Standard CCTV is strictly restricted in enclosed pods due to privacy laws.

Core Dilemma: Conventional management fails to balance real-time occupancy monitoring with privacy protection.



Enclosed Study Pod: High Privacy Sensitivity & High Occupancy Demand

Proposed Solution: Privacy-Preserving Occupancy Monitoring

Edge AI Computing

Real-time on-device video analysis; raw footage never leaves for the cloud, preventing data leaks.

De-identification

Captures only heat signatures, never faces; AI identifies human heat sources to count people.

Remote Oversight

One live dashboard replaces door-to-door checks; rooms left empty too long trigger alerts automatically.

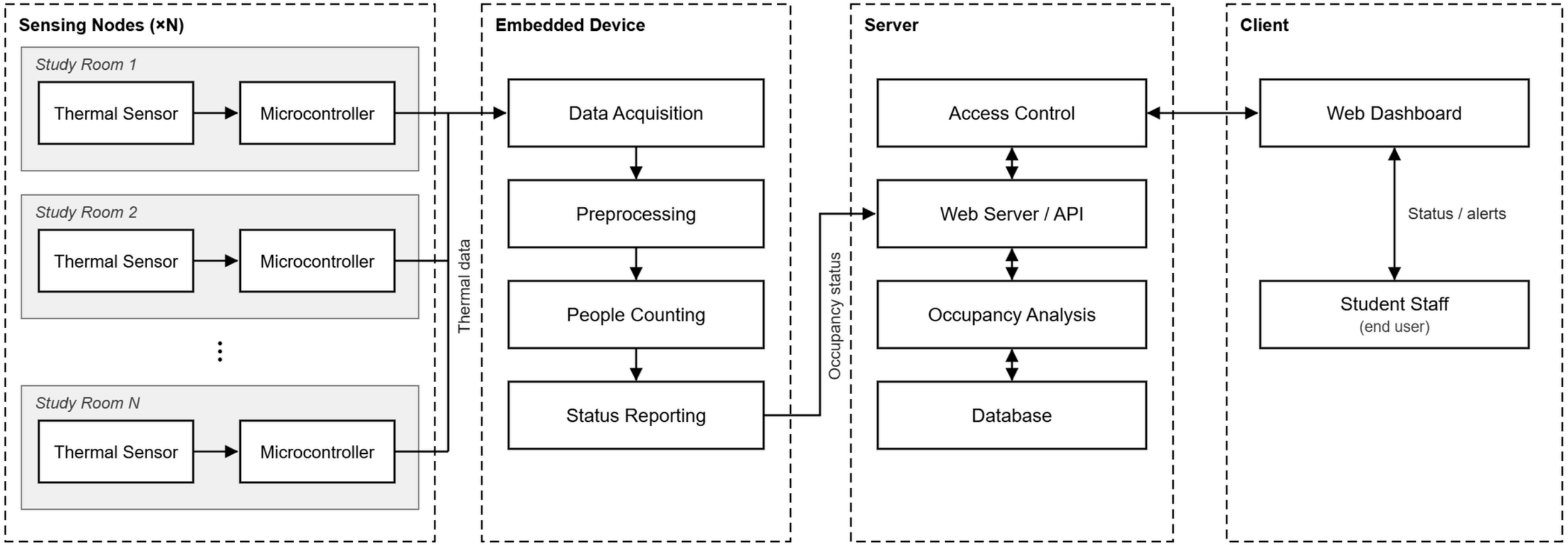


Fig. 1. System block diagram of the MVP.

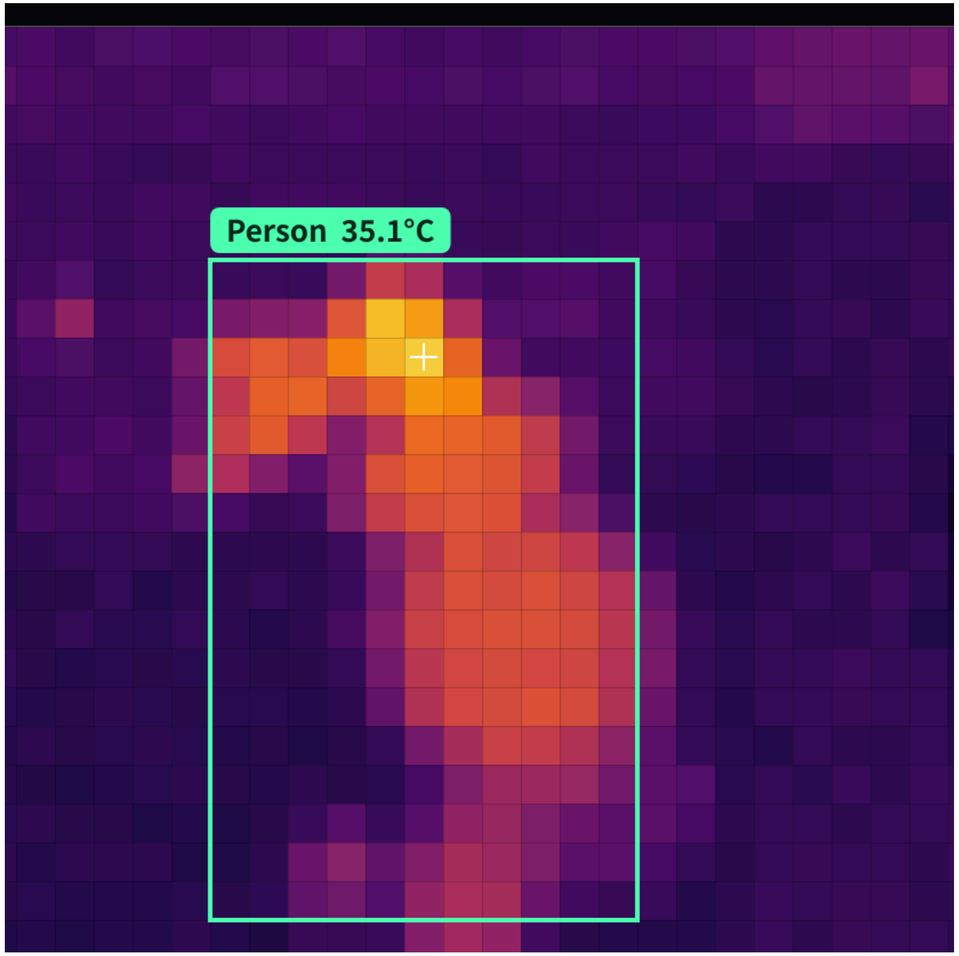
HTTPS://LANYURAIN.GITHUB.IO/RASP_PRO_3D_DEMO/

HTTPS://YOUTU.BE/T9ZU767K1I8

Use Case Demonstration



Corner-mounted



Uses a thermal imaging module to detect human presence while ensuring de-identification.

Project Timeline & Roles





THANK YOU FOR LISTENING

